Opportunistic Multimodal Constraint Satisfaction for Robust Touchscreen Input Disambiguation

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Preprint — April 14, 2026

Abstract

Capacitive touchscreens infer user intent from perturbations of an electrostatic field. This single-modality architecture is fundamentally underdetermined: a raindrop, a vibration artefact, and a deliberate fingertip contact can produce indistinguishable local signatures. We propose a system that resolves this underdetermination through opportunistic cross-modal constraint satisfaction. Under normal conditions the system operates identically to a conventional touchscreen, with no additional overhead or hardware dependency. When primary-signal confidence falls below a dynamically adapted threshold—due to rain, mechanical vibration, or weak-signal contact such as gloved operation—a secondary non-recording wearable device is activated. This device, exemplified as lightweight glasses equipped with optical or infrared sensing, provides real-time hand-motion and postural-context data that is used to validate or reject candidate touch events. The wearable device is defined by principled architectural incapacity: it cannot store imagery, transmit raw sensor data, or produce a visual display, and these constraints are enforced in hardware rather than software. The system additionally supports haptic glove input as a screen-free channel, gated on a postural-context assessment that prevents spurious input when the user's hands are not in a favourable configuration. We describe the system architecture, the formal admissibility model, the operational modes, and the relationship of the approach to prior work in gesture recognition, sensor fusion, and gesture-based text entry. The design is offered as a defensive public disclosure, placing the described methods in the public domain.

Keywords: touchscreen disambiguation, multimodal input, wearable sensing, constraint satisfaction, rain rejection, haptic input, defensive publication.

1. Introduction

The capacitive touchscreen has become the dominant human-computer interface modality for portable devices. Its success rests on a simple and elegant physical principle: a conducting finger perturbs an electrostatic field maintained across a sensing grid, and the location and magnitude of that perturbation is decoded as a touch event. Under controlled conditions this works well. Under real-world conditions it breaks down in ways that have resisted satisfactory engineering solutions for over a decade.

The core difficulty is not one of signal fidelity but of *underdetermination*. From the perspective of the electrostatic grid, a raindrop impacting a glass surface and a fingertip approaching deliberately are, in many configurations, indistinguishable. Both are conducting stimuli of roughly the right size, arriving at the surface with broadly similar temporal profiles. No amount of improvement to the sensing grid itself can resolve this ambiguity, because the grid only has access to the local surface perturbation. It has no view of what caused it.

Existing mitigations—spatial filtering, temporal thresholding, "wet touch" modes, oleophobic coatings, and perimeter moisture detectors—address symptoms rather than causes. They tune the surface measurement to be less sensitive to certain classes of false positive, at the cost of also being less sensitive to some true positives. They cannot recover the missing information, because that information was never available to the surface in the first place.

The solution we propose is to provide that missing information from an independent source. A second sensing modality, geometrically distinct from the screen surface, can observe the user's hand directly. It can answer questions that the surface cannot: is a hand approaching at all? Is a finger extending toward the screen? Does the timing and trajectory of that approach match the candidate touch event detected at the surface? A raindrop cannot answer yes to these questions. A deliberate finger contact typically can.

The challenge is to provide this second modality without introducing the privacy, social, and architectural costs that have caused every previous attempt at "smart glasses" to fail or stall. We argue that this challenge is resolvable through a principle we call *principled incapacity*: the secondary device is defined not only by what it does but by what it is architecturally prevented from doing. A device that cannot record, cannot store, and cannot display is qualitatively different from a general-purpose wearable computer. It occupies the same social category as a keyboard or a mouse—a transient instrument whose operation leaves no durable trace.

A second design principle is *opportunistic activation*. The secondary device is not always on. It is activated only when the primary signal falls below a confidence threshold. Under normal dry conditions with steady hands, the system is indistinguishable from a conventional touchscreen. The secondary device sits idle, consuming negligible power, and imposes no latency. Only when

conditions degrade does it engage.

The result is a system that is robust where current systems are fragile, private where smart glasses have been intrusive, and compatible with existing touchscreen hardware without requiring any modification to the primary device.

The remainder of this paper is organised as follows. Section 2 surveys related work. Section 3 presents the formal model. Section 4 describes the system architecture in detail. Section 5 covers the operational modes. Section 6 addresses haptic glove integration. Section 7 discusses the broader significance of the approach, including its relationship to constraint-based disambiguation in other domains. Section 8 concludes.

2. Related Work

2.1. Touchscreen Filtering and Environmental Compensation

The most direct prior work is the body of engineering addressing adverse-condition touchscreen performance. Manufacturers have introduced glove modes, which raise sensitivity thresholds to respond to non-conductive or partially conductive contacts, and wet-touch modes, which apply additional spatial and temporal filtering to suppress the small, fast, scattered contacts characteristic of rain. Some devices include capacitive moisture detectors at the screen perimeter that can trigger mode changes when water is detected.

These approaches share a common limitation: they operate entirely within the local surface measurement. Raising a sensitivity threshold helps with some false positives but increases false negatives. Lowering it does the reverse. No threshold can simultaneously improve both, because the signal distributions of intentional and non-intentional contacts overlap. The problem is not filter design; it is the absence of an independent discriminating signal.

2.2. Optical and Depth-Based Gesture Recognition

Gesture recognition systems using cameras, depth sensors, or time-of-flight imaging have been extensively explored [1]. These systems can detect hand motion and infer gestures in free space with high accuracy. However, they are almost universally designed as primary input modalities, replacing touch rather than augmenting it. They depend on continuous scene capture, require computational infrastructure for real-time scene understanding, and typically produce persistent video or depth streams.

The present work differs in treating optical sensing as a secondary constraint channel rather than a primary input channel. The wearable device does not need to understand the scene; it only

needs to answer a narrow set of questions about hand position and motion. This dramatically reduces computational requirements and eliminates the need for any persistent data capture.

2.3. Smart Glasses and Augmented Reality

Smart glasses platforms such as Google Glass, Microsoft HoloLens, and Meta Ray-Ban demonstrate that wearable devices can provide useful spatial awareness. However, all of these platforms include persistent recording capabilities, display surfaces, and complex scene-understanding pipelines. The recording capability in particular has generated substantial social resistance: bystanders cannot determine whether they are being observed, and this ambiguity has limited adoption even in applications where recording would be benign.

The device proposed here is distinguished from all of these by the principled absence of recording and display. This is not a policy choice but a hardware constraint, verifiable through attestation. The difference matters socially as much as technically.

2.4. Sensor Fusion in Mobile Devices

Modern mobile devices fuse multiple co-located sensors—accelerometers, gyroscopes, proximity sensors, barometers, and ambient light sensors—to improve orientation estimation, context awareness, and power management. The fusion is entirely internal to the device.

The present work extends sensor fusion across physically separate devices, adding an external wearable viewpoint to the sensing ensemble. This is qualitatively different from intra-device fusion: the wearable provides a perspective from which both the user's hands and the screen surface are simultaneously observable, a viewpoint the device itself cannot achieve.

2.5. Gesture-Based Text Entry and Disambiguation

The Swype keyboard [2] offers an instructive structural parallel. Swype accepts a continuous swipe trajectory across a touchscreen and maps it to a word by projecting the trajectory into a constrained linguistic space: the user's lexicon, weighted by frequency priors and personal usage history. The raw gesture is underdetermined—many words correspond to any given trajectory—and the lexicon provides the constraint that selects among them.

The present system performs an analogous disambiguation at a lower level of the input stack. Rather than asking which symbol a trajectory encodes, it asks whether a surface event encodes any deliberate act at all. The constraint is physical and kinematic rather than linguistic, but the structural principle is identical: raw input is not trusted in isolation and must be brought

into agreement with an independent constraint source before being admitted. We return to this analogy in Section 7.

2.6. Air Typing and Haptic Glove Input

Air-typing and haptic-glove systems encode finger flexion and contact events as symbolic input without requiring physical contact with a surface [3]. These systems have demonstrated reasonable accuracy in controlled settings but share a common weakness: they do not account for postural context. A virtual keyboard projected in front of the user is not useful when the user's hands are at their sides, when the user is walking, or when gaze is directed elsewhere. Without postural gating, glove-mode input generates spurious events whenever the user's hands move for non-input purposes.

The present work integrates haptic glove input within the same constraint framework as touch disambiguation and explicitly gates glove-mode activation on a postural-context assessment derived from the wearable device.

3. Formal Model

3.1. Primary Event Stream

The primary input device produces a stream of candidate touch events:

$$\mathcal{E} = \{\, e_i = (x_i,\, y_i,\, t_i,\, s_i,\, \sigma_i) \}$$

where (x_i, y_i) are screen coordinates, t_i is the detection timestamp, s_i is the integrated signal strength, and $\sigma_i \in [0, 1]$ is a per-event confidence score. The confidence score is produced by a local classifier whose features include signal morphology, contact-area distribution, approach velocity (where derivable from the temporal signal profile), and the current estimated false-positive rate.

3.2. Adaptive Threshold

The confidence monitor maintains a sliding-window estimate of the ambient false-positive rate $\hat{f}(t)$ and reads vibration magnitude $v(t)$ from the device accelerometer. The acceptance threshold is:

$$\tau(t) = \tau_0 + \alpha\, \hat{f}(t) + \beta\, v(t) \tag{1}$$

where τ_0 is a baseline threshold and α, β are tunable weights. Under calm, dry conditions $\tau(t) \approx \tau_0$ and nearly all events satisfy $\sigma_i \geq \tau(t)$, proceeding directly to the host application. Under rain or vibration $\tau(t)$ rises, causing more events to be held pending cross-modal validation.

3.3. Constraint Signal

When activated, the wearable device produces a constraint signal stream:

$$\mathcal{C} = \left\{ c_j = (\mathbf{p}_j,\, \mathbf{v}_j,\, \phi_j,\, t_j,\, \psi_j) \right\}$$

where $\mathbf{p}_j \in \mathbb{R}^3$ is the estimated fingertip position in a shared coordinate frame, $\mathbf{v}_j$ is its velocity vector, $\phi_j \in [0, 1]$ is finger-extension confidence, t_j is the observation timestamp, and $\psi_j \in \{0, 1\}$ is a postural-context flag (defined in Section 6).

3.4. Admissibility

A held candidate event e_i is admitted if and only if $A(e_i, \mathcal{C}) = 1$, where admissibility is the conjunction of three conditions.

Temporal correlation.

$$\exists\, c_j \in \mathcal{C} : |t_i - t_j| \leq \Delta t$$

where Δt is a configurable temporal window (default 120 ms). This condition eliminates environmental events that have no associated hand motion.

Spatial intersection.

$$\left\| \pi(\mathbf{p}_j) - (x_i, y_i) \right\| \leq r$$

where $\pi : \mathbb{R}^3 \to \mathbb{R}^2$ is the coordinate-frame registration transform (Section 4.4) and r is a tolerance radius (default 8 mm). This condition requires that the observed hand position projects onto the screen region of the candidate event.

Motor plausibility. The velocity profile $\mathbf{v}_j$ and the finger-extension sequence leading to c_j must be consistent with voluntary reach-and-contact motion. Criteria include: approach velocity within a characteristic range $[v_{\min}, v_{\max}]$, monotonically decreasing finger-to-screen distance over the preceding window, and absence of lateral scatter inconsistent with directed motion.

Events satisfying all three conditions are promoted to the host application as validated input. Events failing any condition are classified into one of three rejection classes: *environmental noise*

(no temporally proximate constraint observation, e.g. rain or condensation), *vibrational artefact* (constraint observation present but motor plausibility absent), or *non-directed contact* (hand present but approach not directed toward the event location).

4. System Architecture

4.1. Overview

Figure 1 illustrates the high-level architecture. The primary input device generates candidate events, which the confidence monitor evaluates against the adaptive threshold $\tau(t)$. Events meeting the threshold are forwarded directly to the host application. Events below threshold are held and routed to the fusion module, which activates the wearable constraint device and evaluates cross-modal admissibility.

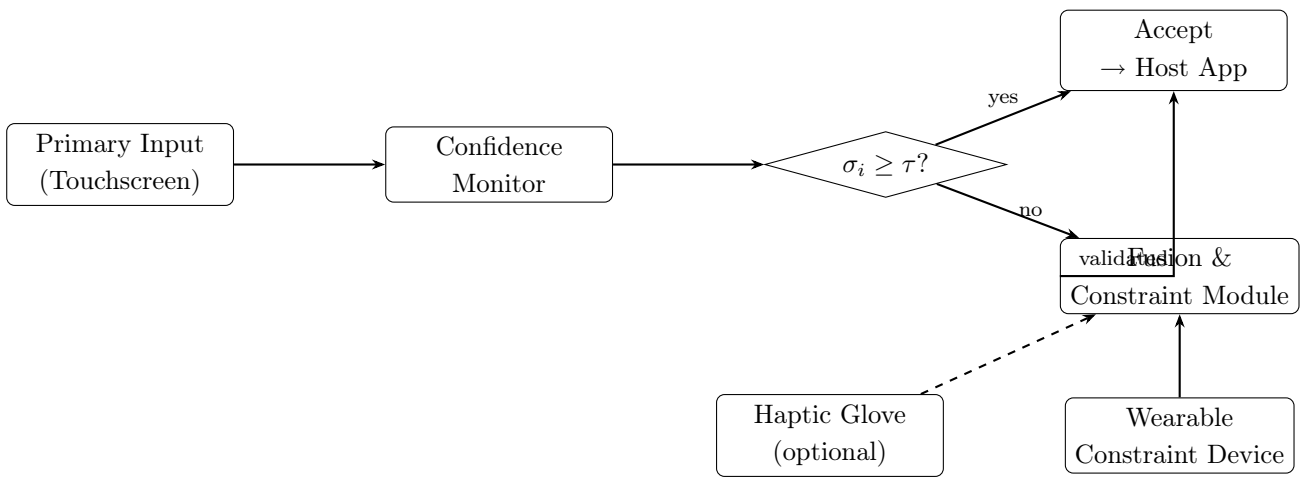

Figure 1: System architecture. The wearable constraint device is activated only when primary-signal confidence falls below the adaptive threshold $\tau(t)$. The haptic glove channel (dashed) is additionally gated on postural context ψ_j.

4.2. Primary Input Device

The primary device requires no hardware modification beyond existing touchscreen designs. It contributes the event stream $\mathcal{E}$, the confidence scores σ_i, and the environmental readings (accelerometer, barometer, optional moisture detector) used by the confidence monitor to compute $\tau(t)$.

4.3. Wearable Constraint Device

The preferred physical embodiment is lightweight glasses. Alternative embodiments include a headband, an ear-mounted unit, or a shoulder-mounted sensor. The requirement common to all embodiments is a viewpoint from which both the user's hands and the device screen are simultaneously observable.

Sensing hardware comprises one or more of: a monocular or stereo RGB or monochrome camera; an infrared illuminator with corresponding IR sensor for low-light operation; a time-of-flight depth sensor; and an inertial measurement unit for head-pose estimation. The device communicates with the host via a low-latency wireless protocol (e.g. Bluetooth Low Energy with a custom constraint-data profile), transmitting only the derived signal $\mathcal{C}$. Raw imagery is never transmitted.

The device is subject to the following hardware constraints, enforced through physical design and verified by a secure-element attestation mechanism:

- No non-volatile storage. All sensor data is processed in working memory and discarded within a bounded window not exceeding 500 milliseconds.
- No display surface, projector, or optical combiner.
- The wireless interface is read-only with respect to raw sensor data; only derived constraint signals conforming to the specified protocol may be transmitted.

These constraints place the device in a qualitatively different category from existing wearable computers. It behaves as a measuring instrument rather than a recording device, analogous to a thermometer or a mouse: it produces a transient output signal and retains no record of the environment.

4.4. Coordinate Frame Registration

Cross-modal spatial correlation requires that the constraint signal $\mathbf{p}_j$ and the screen coordinates (x_i, y_i) be expressed in a common frame. Registration is performed at pairing time via a brief calibration procedure: the user touches a sequence of prompted targets, and the fusion module records simultaneous screen and wearable observations. A rigid-body projection $\pi : \mathbb{R}^3 \to \mathbb{R}^2$ is estimated by least-squares minimisation over the calibration pairs and updated incrementally as validated events accumulate during normal use, accommodating gradual drift from changes in wearable fit or device orientation.

4.5. Fusion and Constraint Evaluation

The fusion module maintains a short event buffer holding candidate events awaiting cross-modal validation. When the wearable is active, each buffered event is evaluated against the admissibility conditions of Section 3. Admitted events are forwarded with a latency not exceeding $\Delta t + d_{\text{comm}}$, where d_{comm} is the wireless round-trip time (target: $\leq 30\,\text{ms}$), giving a worst-case additional latency of approximately 150 ms relative to a direct touch event.

A per-session rejection log is maintained in volatile memory to support adaptive threshold adjustment. This log is not persisted beyond the current session.

A personalised motor model $\mathcal{M}_u$ is maintained in local encrypted storage on the host device and is updated as the user accumulates validated input events. Over time this model refines the motor-plausibility classifier to the specific approach velocities, angles, and inter-event timing of the individual user, reducing false negatives without increasing false positives. The model is device-local and is not transmitted to any external system.

5. Operational Modes

5.1. Normal Mode

The system operates as a conventional touchscreen. The wearable is in low-power idle. All candidate events satisfy $\sigma_i \geq \tau(t)$ and are forwarded directly to the host application. This is the predominant mode under dry, stable conditions.

5.2. Rain and Moisture Rejection Mode

Activated when $\hat{f}(t)$ exceeds a moisture-onset threshold or when a perimeter moisture detector reports wet conditions. The threshold $\tau(t)$ rises according to (1). The wearable is activated. Touch events require cross-modal confirmation. Events lacking a temporally proximate constraint observation are classified as environmental noise and discarded.

The user experience is transparent: deliberate touches continue to register while raindrops are suppressed, at the cost of the validation latency for events that require cross-modal confirmation. In practice, deliberate touches during rain tend to be slower and more deliberate than normal, so the additional latency is largely masked by the approach trajectory itself.

5.3. Vibration Rejection Mode

Activated when accelerometer RMS exceeds a vibration threshold, as occurs on trains, aircraft, or during seismic events. The $\beta\, v(t)$ term in (1) raises the acceptance threshold. The motor-plausibility condition is particularly effective in this mode: vibration-induced contact events lack directed approach trajectories, while intentional touches during vibration are characterised by a bracing-and-reaching motion profile that is detectably different.

5.4. Weak-Signal and Glove Mode

Activated when s_i falls below a glove-detection threshold, indicating a non-conductive or partially conductive contact. The wearable becomes the primary signal source, and contact is inferred from finger trajectory rather than surface perturbation. The electrostatic grid continues to contribute coarse positional data when available but is not required for event validation.

6. Haptic Glove Integration

6.1. Postural Context

The postural-context flag ψ_j encodes whether the user's current body configuration is favourable for deliberate input. The wearable assesses three conditions: (1) hand elevation consistent with active input rather than resting or conversational gesture; (2) head orientation, estimated from the IMU, directed toward the expected input region (a coarse estimate, not requiring eye tracking); and (3) hand and head motion below a stability threshold. $\psi_j = 1$ when all three conditions are met; $\psi_j = 0$ otherwise.

6.2. Glove Input Channel

Haptic gloves encode finger flexion, extension, and contact events as symbolic input without requiring screen contact. Within the present framework, glove-generated events are treated as candidate inputs subject to the same constraint evaluation as touch events, with the addition that the entire glove channel is suspended when $\psi_j = 0$.

This postural gating is the key distinction from prior air-typing systems. It prevents spurious input events from being generated when the user's hands move for non-input purposes: walking, talking with the hands, reaching for objects, or simply resting. The channel activates automatically when the user adopts an input posture and suspends automatically when they leave it, requiring no explicit mode switching by the user.

6.3. Virtual Layout Anchoring

When $\psi_j = 1$, the fusion module anchors a virtual input layout—a keyboard or gesture vocabulary—to the user's physical configuration as derived from the wearable. Repeated finger-motion trajectories are clustered to identify stable symbolic mappings. The layout adapts over time to the user's natural hand position and reach geometry, converging toward a stable, personalised mapping rather than requiring the user to conform to a fixed virtual layout.

7. Discussion

7.1. Constraint Satisfaction as a Unifying Principle

The core contribution of this work is not a specific sensing technology but a reframing of the touch-input problem. Current systems treat touch interpretation as a *classification problem within a single modality*: given the signal at the surface, determine the most likely input event. We propose treating it as a *constraint satisfaction problem across modalities*: an input event is admitted only when it satisfies constraints imposed independently by two or more sensing channels.

This reframing has a well-known analogue in text entry. The Swype gesture keyboard accepts an inherently ambiguous swipe trajectory and resolves it by projecting into a constrained space defined by the user's lexicon. The raw signal is not trusted on its own. It must agree with an independent constraint—language statistics and personal vocabulary—before a specific symbol is committed.

The present system applies the same strategy at the level of physical intent rather than symbolic identity. A touch event is not committed until it agrees with an independent observation of hand motion. The constraint is kinematic and spatial rather than lexical, but the structural principle is identical: raw recurrence (repeated surface perturbations) is resolved by cross-channel constraint into stable output (validated intentional input).

This principle generalises beyond the specific embodiments described here. Any situation in which a primary signal is underdetermined can potentially be stabilised by coupling it to an independent constraint channel, provided that the sources of noise in the primary channel are unlikely to simultaneously satisfy the constraint. Rain satisfies neither temporal correlation with directed hand motion nor motor plausibility. Vibration artefacts satisfy temporal correlation but not motor plausibility. Palm contact satisfies spatial proximity but not directed approach. Each noise source fails at a different point in the admissibility chain.

7.2. Why Previous Smart Glasses Approaches Failed

The history of consumer smart glasses has been one of technically capable devices that failed for social and architectural reasons rather than sensing limitations. The primary issues have been the recording capability and the display: bystanders cannot determine what the device is capturing, and the display creates a visible barrier between the wearer and their social environment.

The device proposed here is architected to avoid both issues by exclusion rather than policy. There is nothing to record because there is no storage. There is nothing to display because there is no display surface. The hardware attestation mechanism makes these absences verifiable rather than merely asserted. A device whose incapacities are demonstrable occupies a different social and regulatory category from one that merely promises not to use capabilities it possesses.

The device is, in this respect, more analogous to a barcode scanner or a motion sensor than to a camera. Its outputs are narrow, structured, and transient. It has no general-purpose sensing capability that could be redirected toward surveillance. This narrowness is a feature rather than a limitation.

7.3. Graceful Degradation

A design requirement not always emphasised in multimodal systems is graceful degradation: the system should not perform worse when auxiliary devices are absent than a conventional system without them. The present design satisfies this requirement by construction. When the wearable is unavailable, the system reverts to conventional single-surface inference. When the wearable is idle, it imposes no latency or power overhead. The auxiliary channel is strictly additive.

7.4. Privacy

The system transmits no biometric data in the conventional sense: no face images, no gaze direction, no voice. The constraint signal encodes only coarse hand position and velocity, which are not individually identifying. The personalised motor model is stored locally and not transmitted. The system does not require network connectivity to function.

7.5. Limitations and Future Work

The primary limitation of the current design is the calibration requirement. The coordinate-frame registration between the wearable and the screen requires an explicit calibration step at pairing time and periodic re-calibration as device position or wearable fit changes. Future work could address this through continuous self-calibration using validated touch events as ground

truth, reducing or eliminating the explicit calibration step.

A second limitation is the validation latency. The worst-case additional latency of approximately 150 ms is imperceptible for many interactions but may be noticeable for rapid tap sequences. Future work could reduce this by predictive pre-activation of the wearable when environmental conditions suggest imminent degradation, reducing the latency of the first event in a degraded-condition sequence.

The motor-plausibility classifier described here is relatively simple. Richer models trained on larger datasets of intentional versus non-intentional contacts across a range of environmental conditions would improve accuracy, particularly in novel adverse conditions not well represented in the training data.

8. Conclusion

We have described a system for robust touchscreen input disambiguation based on opportunistic cross-modal constraint satisfaction. The system introduces a secondary non-recording wearable device that provides real-time hand-motion constraint data, activated only when primary-signal confidence is insufficient. Input events are admitted only when they satisfy temporal, spatial, and motor-plausibility conditions across both sensing modalities simultaneously. The wearable device is defined by principled architectural incapacity—it cannot record, store, or display—placing it in a social and regulatory category distinct from existing smart glasses systems. A haptic glove input channel is integrated within the same constraint framework, gated on a postural-context assessment that prevents spurious input outside deliberate input postures.

The approach reframes touch input from a single-modality classification problem to a cross-modal constraint satisfaction problem, a reframing with structural analogues in gesture-based text entry and other domains where raw signals are underdetermined relative to intended outputs.

This paper is offered as a defensive public disclosure. The methods described are hereby placed in the public domain. No patent protection is claimed, and the authors encourage free implementation and extension of these ideas.

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This work is released as a defensive public disclosure. The described methods are placed unconditionally in the public domain as of the date of first publication. No patent application has been filed. The authors assert no proprietary rights over the techniques described herein and encourage free use, implementation, and extension.